夏松

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教育经历

博士研究生，电子与电气工程学院，南洋理工大学 2022 – 2026(expected)

指导教师：Professor Jiang Xudong

Rapid-Rich Object Search (ROSE) Lab

硕士研究生，电子与电气工程学院，南洋理工大学 2021 - 2022

指导教师： Professor Yap Kim Hui

论文题目: Certified Defense and Robust AI

均分: 4.80/5.00

本科， 电子信息学院，武汉大学 2016 - 2020

指导教师：江昊教授

论文题目: 基于强化学习的对抗样本生成和防御

均分: 84/100

研究方向

- 可信人工智能
- 对抗鲁棒性
- 多模态大模型安全

论文成果

- **Song Xia**, Yi Yu, Wenhan Yang, Meiwen Ding, Zhuo Chen, Lingyu Duan, Alex C. Kot, Xudong Jiang. Theoretical Insights in Model Inversion Robustness and Conditional Entropy Maximization for Collaborative Inference Systems. The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR Highlight), 2025.

- **Song Xia**, Wenhan Yang, Yi Yu, Xun Lin, Henghui Ding, Lingyu Duan, Xudong Jiang. Transferable Adversarial Attacks on SAM and Its Downstream Models. In The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

- **Song Xia**, Yi Yu, Xudong Jiang, Henghui Ding. Mitigating the Curse of Dimensionality for Certified Robustness via Dual Randomized Smoothing. In ICLR 2024

- **Song Xia**, Yi Yu, Henghui Ding, Wenhan Yang, Shifei Liu, Alex C Kot, Xudong Jiang. Open-set Anomaly Segmentation in Complex Scenarios, In arXiv, 2025 (Under Review, IEEE TIP major revision).

- **Song Xia**, Meiwen Ding, Chenqi Kong, Wenhan Yang, Xudong Jiang. Feature-Space Adversarial Robustness Certification for Multimodal Large Language Models. In arXiv preprint arXiv:2601.16200. (under review by ICML 2026)

- Meiwen Ding, **Song Xia***, Chenqi Kong, Xudong Jiang, Adversarial Prompt Injection Attack on Multimodal Large Language Models. (Under review by ICML 2026)

- Meiwen Ding, **Song Xia***, Yi Yu, Shuting He, Xudong Jiang, Transferable Adversarial Attack on Referring Video Object Segmentation. (Under Review, IEEE TIFS major revision)

- Joy Dhar, **Song Xia**, Manish Kumar Pandey, Maryam Haghighat, Azadeh Alavi, Ferdous Sohel, Wenyu Zhang, Nayyar Zaidi**,** Certified vs. Empirical Adversarial Robustness via Hybrid Convolutions with

Attention Stochasticity. In ICLR 2026.

- Yi Yu, **Song Xia**, Xun Lin, Chenqi Kong, Wenhan Yang, Shijian Lu, Yap-Peng Tan, Alex C Kot. Towards Model Resistant to Transferable Adversarial Examples via Trigger Activation. In IEEE TIFS, 2025.

- Yi Yu, **Song Xia**, Xun Lin, Wenhan Yang, Shijian Lu, Yap-peng Tan, Alex Kot. Backdoor Attacks against No-Reference Image Quality Assessment Models via A Scalable Trigger. In 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

- Yi Yu, Yufei Wang, **Song Xia**, Wenhan Yang, Shijian Lu, Yap-Peng Tan, Alex C Kot. Purify Unlearnable Examples via Rate-Constrained Variational Autoencoders. In Proceedings of the 41st International Conference on Machine Learning (ICML), 2024.

参与项目

- Towards More Trustworthy Generative AI through Robustness Enhancement and Hallucination Reduction. (新加坡IMDA，金额130W新币.)

- Towards the Cohesive Safeguard Architecture and Comprehensive Robustness Evaluation for Multimodal Large Language Model Agents. (新加坡MOE Tier2， 金额90W新币)